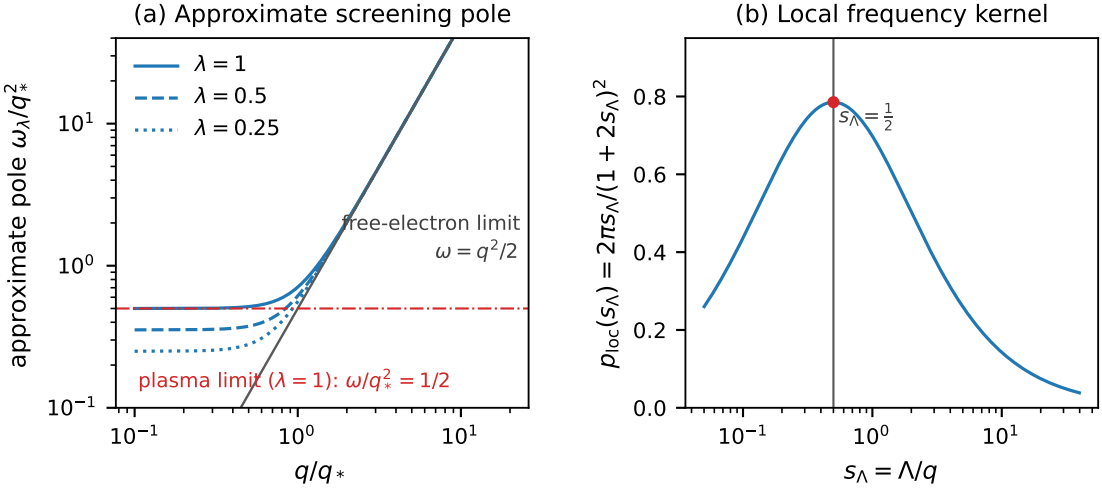
(a) Approximate screening pole
$\lambda = 1$
$\lambda = 0.5$
$\lambda = 0.25$
free-electron limit
$\omega = q^2/2$
plasma limit ($\lambda = 1$): $\omega/q_*^2 = 1/2$
approximate pole ω_λ/q_*^2
q/q_*
(b) Local frequency kernel
$p_{\mathrm{loc}}(s_\Lambda) = 2\pi s_\Lambda/(1+2s_\Lambda)^2$
$s_\Lambda = \frac{1}{2}$
$s_\Lambda = \Lambda/q$